On the Effectiveness of Invisible Backdoor Attacks in Federated Learning

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01 Context & approach

The challenge

Invisible triggers must survive aggregation with honest updates. This study measures when existing backdoor attacks remain effective in federated learning and whether low-activation placement improves success without making the attack easier to detect.

Our contribution

Compare four invisible attack families with visible BadNet triggers.
Measure attack success, clean accuracy, visual distance, and detection scores.
Propose and evaluate low-activation trigger placement for LSB attacks.

1 Run a controlled comparison

Honest clients train normally; attackers use clean and poisoned samples. Aggregate updates across repeated federated rounds.

2 Vary attack conditions

Compare LSB, WaNet, Encoder, and LIRA with BadNet. Vary attacker prevalence and evaluate selected server defenses.

3 Test low-activation placement

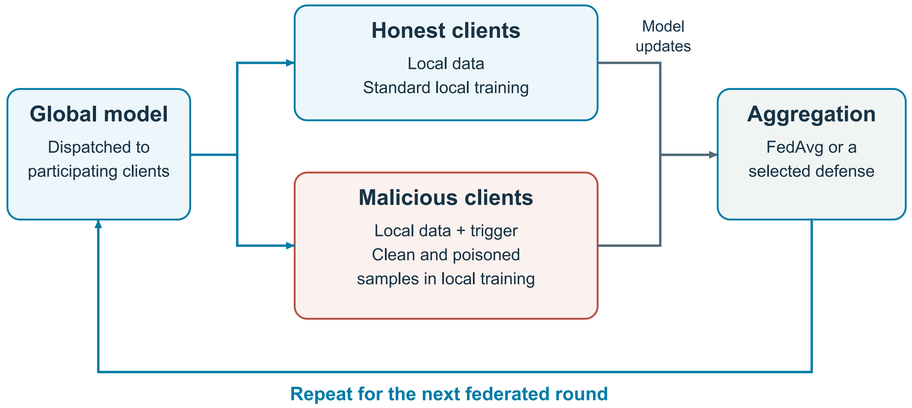
Place LSB triggers in low-activation regions, then compare attack success and detection scores with unmasked and high-activation controls.

Scope

Experiments use ResNet-18 and small image datasets; run count and the meaning of ± are unspecified. Placement evidence covers LSB. Greater visual similarity does not imply evasion of model-level detection.

02 Method

Evaluate subtle triggers, attacker prevalence, and the effectiveness-detectability trade-off.



Core formulation

$$\mathcal{L}_{\text{attack}} = \mathcal{L}_{\text{clean}} + \mathcal{L}_{\text{poison}\to t}$$

The two cross-entropy terms fit clean labels and poisoned samples labelled as target class t. Honest updates compete with this objective.

$$\text{ASR}(\%) = 100\,\frac{N_{\text{triggered}\to t}}{N_{\text{triggered}}}$$

$N_{\text{triggered}}$: poisoned test samples excluding the target class; the numerator counts those predicted as target t. ASR measures attack success.

Paper

Code

03 Results

88.59%
Encoder attack success
20% attackers · CIFAR-10

+19.80 pp
LSB700 ASR difference
Low minus unmasked placement

Attacker prevalence changes attack success

Attack	Clean acc., 10%	ASR, 10%	ASR, 20%
BadNet	84.00 ± 0.31	94.96 ± 0.81	97.52 ± 0.40
LSB	83.37 ± 0.33	41.59 ± 6.19	65.43 ± 3.93
WaNet	83.33 ± 0.31	44.41 ± 3.50	82.05 ± 3.59
Encoder	83.44 ± 0.27	29.44 ± 5.90	88.59 ± 4.01
LIRA	83.71 ± 0.41	10.52 ± 1.22	41.57 ± 4.66

CIFAR-10 · ResNet-18 · FedAvg · 10 clients · 100 rounds · Dir(0.5). Poison rate 0.5; target class 5. Columns give attacker fraction; reported ± is undefined.

Low-activation placement has a cost

LSB700 placement	ASR (%)	LPIPS	Anomaly index
Unmasked	41.59 ± 6.19	7.83e-5	1.61
Low (Ours)	**61.39 ± 4.47**	**2.25e-4**	**3.04**
High	36.89 ± 4.26	3.75e-4	2.01

LSB700 · CIFAR-10 · 10% attackers. Lower LPIPS means closer images; Neural Cleanse flags anomaly above 2. “Low” is the proposed placement.

Takeaway

Encoder reaches **88.59% ASR with 20% attackers**, showing the importance of attacker prevalence. Proposed **low-activation LSB700 placement** gains 19.80 points over unmasked placement, but raises Neural Cleanse’s anomaly index from 1.61 to 3.04.